

2nd Year Electronics & Telecomm Engg. (Section A & B)
3ETRC3 Data Structure

Time: 70 min

Max Marks: 20

Note: 1. Attempt any four questions. All questions carry equal marks.

2. Write the output with sample input while answering programming question

Q.1 Write C program to find the number = 23 in the following data set using Binary Search algorithm.
Data[] = {5, 7, 9, 21, 23, 35, 78, 89, 93, 98}

Q.2 Illustrate the quick sort algorithm to sort the following data set.
Arr[] = {30, 52, 29, 87, 63, 27, 19, 54}

Q.3 What are the various applications of stack and queue. Explain in brief.

Q.4 Convert the following expression to Polish and Reverse Polish notation:
Expression: $a + (b * c + d) / e$

Q.5 How Circular queue overcomes limitation of linear queue? Write the algorithm for Insertion and Deletion in a circular queue